

Topics in Error-Correcting Codes

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Lecture 7: Tree Codes (September 30, 2026)

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Recap

In the last two lectures we defined the notion of a tree code, and showed how to use a tree code to do interactive coding. In this lecture we present a tree code construction due to Schulman [1]. We first recall the definition of a tree code.

Definition 1. A tree code, T , is an online code with,

$$T : \{0, 1\}^* \rightarrow \Sigma^*,$$

such that the following conditions hold:

- $|x| = |y| \implies |T(x)| = |T(y)|$.
- For every $x \in \{0, 1\}^*$, $T(x)$ is the prefix of $T(x\|0)$ and $T(x\|1)$.
- For every $t \geq 1$ and distinct $x, x' \in \{0, 1\}^t$, letting s be the first position at which they differ,

$$\Delta(T(x), T(x')) \geq \delta c(t - s + 1),$$

where Δ denotes Hamming distance measured in bits.

- There exists a constant $c \in \mathbb{N}$ such that $|\Sigma| = 2^c$.

Construction of Tree Codes

We still need to argue that tree codes exist. At first, it is not immediately clear that such an object exists. For example, a natural first attempt is to label every edge of the complete binary tree of depth N independently and uniformly with a string in $\mathbb{F}_2^c$. The encoding of an input is the concatenation of the edge labels on its root-to-vertex path.

Random labeling does not give a tree code. It turns out that random labeling does not give a good tree code.¹ To see why, consider any internal vertex v . Its two outgoing edges have equal labels with probability 2^{-c} . If these labels agree, then this violates the definition of a tree code (for every $\delta > 0$).

¹ This is in contrast to the fact that a random error correcting code is a good code with high probability.

There are $2^N - 1$ internal vertices, and the collision events at different vertices are independent. Consequently,

$$\Pr[T \text{ is a tree code}] \leq (1 - 2^{-c})^{2^N - 1} = \text{negl}(N).$$

The issue is that there are exponentially many independent opportunities for a short obstruction to occur. As opposed to an error correcting code that only needs to create distance over the full codeword, a tree code must create distance for every separate paths that diverged only one step ago.

Would making sibling labels distinct fix the problem? No. Suppose that each vertex independently receives a uniformly random ordered pair of distinct labels. Fix an integer $\ell > 1/\delta$. In a depth- ℓ subtree, compare the paths 0^ℓ and $10^{\ell-1}$. Their first labels differ, but their remaining $\ell - 1$ pairs of labels can all agree. This event has probability

$$2^{-c(\ell-1)} > 0.$$

On this event the bit distance is at most c , whereas the required distance is $\delta c \ell > c$. At depth $N - \ell$, there are $2^{N-\ell}$ disjoint subtrees in which this obstruction can occur independently. Thus this modified random labeling also fails with probability tending to one.

Lesson: Fixing one particular short obstruction does not address the repeated independent opportunities for a slightly longer obstruction.

Idea: Use Linearity

For a linear code, the distance between two codewords becomes the weight of a single codeword:

$$\Delta(T(x), T(x')) = \text{wt}(T(x \oplus x')).$$

Thus, for linear codes, only the input difference matters! Since T is required to be an online encoding, we must use a lower triangular matrix for the encoding.

It turns out that if we take a random lower triangular matrix that is **Toeplitz**, meaning the coefficients repeat along diagonals, then the resulting encoding is **time invariant** in the following sense: For every vector $z \in \{0, 1\}^t$ such that $z_s = 1$ and $z_1 = \dots = z_{s-1} = 0$, it holds that

$$T(z) = T(0^{s-1} \| z_s, \dots, z_t) = 0^{c \cdot (s-1)} \| T(z_s, \dots, z_t), \quad (1)$$

where we think of $T(z_s, \dots, z_t)$ as a bit string in $\{0, 1\}^{c \cdot (t-s+1)}$.

As we show below a random linear code which is time invariant, in the sense of Equation (1), is a good tree code with high probability (for a sufficiently large constant c , which depends on the desired distance δ).

The Linear Construction

Choose coefficient blocks uniformly at random

$$a_0, a_1, \dots, a_{N-1} \leftarrow \mathbb{F}_2^c.$$

At time t , output

$$y_t = \sum_{j=1}^t a_{t-j} x_j = a_0 x_t + a_1 x_{t-1} + \dots + a_{t-1} x_1. \quad (2)$$

All operations are over $\mathbb{F}_2$: multiplication by a bit either keeps or zeros a block, and addition is bitwise XOR. In particular,

$$\begin{aligned} y_1 &= a_0 x_1, \\ y_2 &= a_1 x_1 + a_0 x_2, \\ y_3 &= a_2 x_1 + a_1 x_2 + a_0 x_3. \end{aligned}$$

Equivalently,

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} a_0 & 0 & 0 & \dots & 0 \\ a_1 & a_0 & 0 & \dots & 0 \\ a_2 & a_1 & a_0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{N-1} & a_{N-2} & a_{N-3} & \dots & a_0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_N \end{pmatrix}.$$

The matrix is lower triangular, which gives online encoding. In addition, the matrix is Toeplitz, meaning that it is constant on each diagonal. This implies that the construction is time invariant, i.e., for every $z \in \{0, 1\}^t$ such that $z_s = 1$ and $z_1 = \dots = z_{s-1} = 0$,

$$T(z) = 0^{c \cdot (s-1)} \| T(z_s, \dots, z_t).$$

Thus, the difference between the labels of two siblings is always a_0 , regardless of the vertex or its depth. Thus, satisfying $\text{wt}(a_0) \geq \delta c$ takes care of every sibling pair simultaneously. The same phenomenon holds for longer divergences. For example, every pair whose divergent suffix has length two is governed by just one of

$$T(1, 0) = (a_0, a_1) \quad T(1, 1) = (a_0, a_0 + a_1)$$

If both of these vectors have Hamming weight at least $\delta \cdot 2c$ then this takes care of all such pairs everywhere in the tree. More generally, the following claim holds:

Claim 1 (Only inputs starting with 1 need to be checked). The encoder in (2) is a tree code of relative distance δ if and only if, for every $1 \leq \ell \leq N$ and every $z \in \mathbb{F}_2^\ell$ with $z_1 = 1$,

$$\text{wt}(T(z)) \geq \delta c \ell. \quad (3)$$

Proof. Suppose (3) holds. Fix any $t \in [N]$ and any two paths $x, x' \in \mathbb{F}_2^t$. Denote by $s \in [t]$ the first point in which these two vectors diverge; i.e., $x_s \neq x'_s$ and $x_i = x'_i$ for every $i \in [s-1]$. Let

$$\ell = t - s + 1, \quad z = (x_s + x'_s, \dots, x_t + x'_t).$$

Then $z_1 = 1$ and $x + x' = 0^{s-1} \| z$. By linearity and time-invariance,

$$\begin{aligned} T(x) + T(x') &= T(x + x') \\ &= T(0^{s-1} \| z) \\ &= 0^{c \cdot (s-1)} \| T(z). \end{aligned}$$

Therefore

$$\Delta(T(x), T(x')) = \text{wt}(T(z)) \geq \delta c \ell = \delta c(t - s + 1).$$

Conversely, apply the tree-code condition to the pair $z, 0^\ell$, whose first difference is at position 1. $\square$

This is the main conceptual step. Linearity removes the common history; time-invariance removes the starting time. At length ℓ , only the $2^{\ell-1}$ difference patterns beginning with 1 remain. There is no additional union bound over the vertices of the tree or over the times at which divergence begins.

A random coefficient sequence works

Theorem 2 (Linear tree-code existence). *For any $0 < \delta < 1/2$ and any large enough constant $c = O_\delta(1)$,² the following holds: For every depth N , if $a_0, \dots, a_{N-1}$ are independent and uniform in $\mathbb{F}_2^c$, then the encoder in (2) is a tree code of relative bit distance $\geq \delta$ and rate $1/c$ with probability greater than $1/2$.*

Proof. Fix any $N \in \mathbb{N}$. To prove Theorem 2 we need to prove that

$$\Pr[\forall t \in [N] \forall z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) \geq \delta c t] \geq \frac{1}{2}$$

By the union bound

$$\begin{aligned} &\Pr[\forall t \in [N] \forall z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) \geq \delta c t] = \\ &1 - \Pr[\exists t \in [N] \exists z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) < \delta c t] \geq \\ &1 - \sum_{t=1}^N \sum_{z \in \{0, 1\}^{t-1}} \Pr[\text{wt}(T(1 \| z)) < \delta c t] \end{aligned}$$

Note that it suffices to prove that for every $t \in [N]$ and every $z \in \{0, 1\}^{t-1}$

$$\Pr[\text{wt}(T(1 \| z)) < \delta c t] \leq 2^{-2t}. \quad (5)$$

This is the case since Equation (5) implies that

$$\sum_{t=1}^N 2^{t-1} \Pr[\text{wt}(T(1 \| z)) < \delta c t] \leq \sum_{t=1}^N 2^{t-1} \cdot 2^{-2t} \leq \sum_{t=1}^N 2^{-(t+1)} \leq \frac{1}{2},$$

² Specifically, we need c to satisfy

$$c(1 - H(\delta)) \geq 2, \quad (4)$$

where H is the entropy function defined by $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$.

as desired. Thus, it remains to prove Equation (5). To this end, note that $T(1||z)$ is a truly uniform vector in $(\mathbb{F}_2^c)^t$. To see this, fix $w = 1||z$. Its i th output block is

$$y_i = a_{i-1} + \sum_{j=2}^i a_{i-j} w_j.$$

Thus, conditioned on $a_0, \dots, a_{i-2}$, the block y_i is uniform, since a_{i-1} is fresh and uniform. Consequently, the output blocks are jointly uniform and independent.

Thus

$$\begin{aligned} \Pr[\text{wt}(T(1||z)) < \delta \cdot c \cdot t] &= 2^{-c \cdot t} \sum_{\substack{r \in \mathbb{Z} \\ 0 \leq r < \delta \cdot c \cdot t}} \binom{c \cdot t}{r} \\ &\leq 2^{-c \cdot t} \cdot 2^{c \cdot t \cdot H(\delta)} \\ &= 2^{-c \cdot t \cdot (1 - H(\delta))} \\ &\leq 2^{-2t}, \end{aligned}$$

where the second inequality follows from the entropy bound assuming $\delta < \frac{1}{2}$, and the last inequality follows from the definition of c (see Equation (4)). $\square$

Takeaway. An independently random labeling gives exponentially many independent opportunities for the same short obstruction. The linear Toeplitz construction deliberately correlates the labels:

$$\begin{array}{ll} \text{linearity} & \implies \text{only the input difference matters,} \\ \text{time-invariance} & \implies \text{the starting time does not matter.} \end{array}$$

What have we constructed?

A single infinite tree code. Sample an infinite independent sequence $a_0, a_1, \dots$. The same countable union bound gives

$$\Pr[\text{some finite-length requirement fails}] \leq \sum_{\ell=1}^{\infty} 2^{-\ell-1} = \frac{1}{2}.$$

Hence a single coefficient sequence satisfies every finite-depth requirement with probability at least $1/2$. This establishes compatible infinite-depth existence directly, rather than choosing unrelated good codes at different depths.

Efficient encoding, given the coefficients. A depth- N code has a cN -bit description. Given the coefficient blocks, the output at time t is computable in time $O(t \cdot c^2)$. This time bound is for the evaluation of a specified encoder; it does not give an efficient deterministic method for finding good coefficients.

Finding and certifying good coefficients. So far, we showed that we can efficiently sample coefficients which, with probability at least $1/2$, define a tree code. However, this argument does not provide an efficient way to check that the sampled coefficients are good. There is an exponential-time deterministic construction: enumerate the 2^{cN} coefficient sequences and test (3) on the

$$\sum_{\ell=1}^N 2^{\ell-1} = 2^N - 1$$

patterns that begin with 1. For fixed δ and c , this takes $2^{O(N)}$ time. In particular, a depth- $O(\log n)$ tree code can be found in $\text{poly}(n)$ time by this search.

We note that the distinction between existence and explicit efficient construction is central to tree codes. As of today, constructing an explicit asymptotically good constant-alphabet tree codes remain an open problem.

Finally, we mention that argument does not supply an efficient decoder for noisy encodings. Next lecture we will see a different interactive coding scheme with efficient encoding and decoding.

Summary So far we saw that we can sample at random coefficients which with probability $\geq \frac{1}{2}$ define a tree code, and this tree code can be use to for interactive-coding. However, the resulting simulation is not computationally efficient, since the tree code construction is not efficiently decodable. In addition, we certify, or efficiently check, that the sampled coefficients indeed correspond to a tree code.

References

- [1] Leonard J. Schulman. Deterministic coding for interactive communication. In *Proceedings of the 25th Annual ACM Symposium on Theory of Computing (STOC)*, pages 747–756. ACM, 1993.